AN IMPURITY OVER THE HENSELIZATION GIVES ONE OVER AN ELEMENTARY ETALE NEIGHBOURHOOD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The henselization as a limit of affine elementary etale neighbourhoods, and descent of impurities along a filtered limit. More on Morphisms, label more-morphisms-lemma-describe-henselization (proof line 4761); Same chapter flat.tex, label lemma-impure-limit (proof line 4763, claimed as S14-algebra-impure-limit). The first presents the henselization as a filtered limit of the elementary etale neighbourhoods, and the second descends the impurity to some stage of that limit.

Source reference. Original lemma 05IZ: sources/stacks-flat/pinned-original-source.tex, source lines 4573–4573. Navigation pointer for \ref{lemma-impure-limit} used inside the counted span (line 4763); the argument that consumes it is retained in the span.

Source reference. Original lemma 05KS: sources/stacks-chow/tags.txt, source lines 7211–7211. Navigation pointer for \ref{more-morphisms-lemma-describe-henselization} used inside the counted span (line 4761); the argument that consumes it is retained in the span.

Source reference. Original reference 05FW: sources/stacks-flat/pinned-original-source.tex, source lines 4473–4473. Navigation pointer for \ref{situation-pre-pure} used inside the counted span (line 4745); the argument that consumes it is retained in the span.

Lemma 0.1. *In Situation 05FW. If there exists an impurity $(S^h \to S, s' \rightsquigarrow s, \xi)$ of $\mathcal{F}$ above s then there exists an impurity $(T \to S, t' \rightsquigarrow t, \xi)$ of $\mathcal{F}$ above s where $(T, t) \to (S, s)$ is an elementary étale neighbourhood.*

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Proof. We may replace S by an affine neighbourhood of s. Say $S = \mathrm{Spec}(A)$ and s corresponds to the prime $\mathfrak{p} \subset A$. Then $\mathcal{O}_{S,s}^h = \mathrm{colim}_{(T,t)} \Gamma(T, \mathcal{O}_T)$ where the limit is over the opposite of the cofiltered category of affine elementary étale neighbourhoods (T, t) of (S, s), see More on Morphisms, Lemma 05KS and its proof. Hence $S^h = \lim_i T_i$ and we win by Lemma 05IZ. $\square$